Analysis of Interpretable Systems-Level Prioritization of Pan-Cancer Driver Genes Using a Multi-View Transformer-Graph Convolutional Network (TGCN)

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27/08/2026

1 Introduction

1.1 Biological problem

Cancer develops through somatic mutation that disrupts normal cellular processes. A major challenge in cancer genomics is distinguishing driver genes: which contribute functionally to tumor development, from passenger genes: which are altered without contributing directly to cancer progression [1, 3].

While large scale sequencing efforts such as The Cancer Genome Atlas have catalogued mutations across thousands of tumors, most mutated genes are passengers with no functional role in cancer progression [2, 4]. Because cancer is a systems-level disease, individual genes cannot always be interpreted independently. Driver genes may occupy important positions in molecular interaction networks, participate in functional modules, and regulate pathways involved in processes such as proliferation, DNA repair, apoptosis, and signal transduction.

Therefore, computational methods that combine molecular features with biological network information can provide a more comprehensive strategy for candidate driver-gene prioritization.

1.2 Defining the question

> *Can a multi-view Transformer–Graph Convolutional Network (TGCN) prioritize biologically meaningful cancer driver genes, and do the prioritized genes exhibit coherent network and functional characteristics?*

This question is addressed in four stages:

1. Can TGCN distinguish known cancer driver genes from non-driver genes?
2. Do highly ranked TGCN genes occupy important positions in the global interaction network?
3. Do the prioritized genes cluster into biologically relevant network modules?
4. Are the prioritized genes enriched in biologically meaningful pathways?

1.3 Overall Analysis Strategy

The complete analysis follows:

$$\boxed{\text{Biological features + Network information} \rightarrow \text{TGCN} \rightarrow \text{Gene scores} \rightarrow \text{Top 100 genes}}$$

$$\boxed{\text{Top 100} \rightarrow \begin{cases} \text{Network centrality} \\ \text{Network perturbation} \\ \text{Community analysis} \\ \text{Pathway enrichment} \end{cases}}$$

The downstream analyses are complementary. Centrality describes *where* the genes are positioned in the network, community detection describes *which network modules* they belong to, and pathway enrichment describes *which biological processes* are represented.

2 Specifications of Data

2.1 Data Sources and Acquisition

All data used in this project were obtained from the publicly available TGCN repository, which provides unprocessed multi-omics data for pan-cancer driver gene identification. The original data sources documented in the TGCN paper are:

- **TCGA Pan-Cancer Data**: Somatic mutation frequencies, differential DNA methylation, and differential gene expression from The Cancer Genome Atlas (TCGA) Pan-Cancer Analysis of Whole Genomes (PCAWG) consortium across 16 cancer types
- **miRTarBase**: miRNA-gene regulation features for 1,884 genes and 826 miRNAs
- **STRING database (v11.5)**: Protein-protein interaction network with confidence threshold ≥ 0.850
- **Gene Ontology (GO)**: Gene semantic similarity network constructed from GO annotations using the R package GOSemSim [18]
- **KEGG Pathways**: Human cancer pathways for pathway co-occurrence network
- **Cancer Driver Catalogs**: Positive driver genes integrated from NCG v6.0 [11], COSMIC Cancer Gene Census (CGC) [4], and DigSEE
- **Non-driver genes**: Genes not annotated as drivers or disease genes in NCG, COSMIC, OMIM, or KEGG cancer pathways

2.2 Specifications of Data Used for This Project

Following are the final specifications of data used for this project:

- **Biological features**: 51-dimensional feature matrix per gene
  - Mutation rates (somatic mutations across 16 cancer types)
  - Differential DNA methylation
  - Differential gene expression
  - miRNA-gene regulation features
  - Overall somatic mutation frequency
- **PPI network**: STRING database with 10,743 genes and 103,087 edges
- **GO network**: Gene semantic similarity network from GO annotations
- **Pathway network**: KEGG human cancer pathway co-occurrence network
- **Node2Vec embeddings**: 16-dimensional structural embeddings for each network [17]
- **Total input features**: 67 dimensions (51 biological + 16 structural)
- **Driver genes (positive)**: 796 genes
- **Non-driver genes (negative)**: 2,187 genes
- **Genes with valid out-of-fold predictions**: 1,933 genes
- **Cross-validation**: 10 repeats $\times$ 5-fold (50 total folds)
- **Training epochs**: 550 per fold
- **Top prioritized genes for analysis**: Top 100 by TGCN score
- **Candidate genes for analysis**: Top 100 with label = 0 (not in reference driver set)

2.3 Data Preprocessing Pipeline

The following preprocessing steps were performed on the data obtained from the TGCN repository:

2.3.1 Feature Extraction and Gene Filtering

The original feature matrix contained 13,472 genes with 51 biological features. The PPI network was filtered to retain only genes present in both the feature matrix and the interaction network:

1. Loaded the STRING PPI network (10,743 nodes)
2. Identified intersection between feature matrix genes (13,472) and PPI network genes (10,743)

3. Filtered PPI edges to retain only interactions where both source and target genes are in the intersection

4. Extracted biological features for the 10,743 filtered genes

5. Saved filtered feature matrix (10,743 genes × 51 features) and filtered PPI network (103,087 edges)

2.3.2 Node2Vec Structural Embeddings

Three separate Node2Vec models were trained to learn 16-dimensional structural embeddings:

- **PPI network**: `embedding_dim=16`, `walk_length=80`, `context_size=5`, `walks_per_node=10`, `num_negative_samples=1`, $p = 1$, $q = 1$
- **GO semantic similarity network**: Same hyperparameters, `num_nodes=10,743`
- **Pathway co-occurrence network**: Same hyperparameters, `num_nodes=10,743`

Each model was trained for 1,000 epochs with batch size 128 and learning rate 0.0001 using SparseAdam optimizer.

2.3.3 Feature Concatenation

The final input features were constructed by concatenating biological features with structural embeddings:

$$\mathbf{X}_{\text{final}} = [\mathbf{X}_{\text{biological}} \,\|\, \mathbf{X}_{\text{Node2Vec}}] \tag{1}$$

where $\mathbf{X}_{\text{biological}} \in R^{10743\times51}$ and $\mathbf{X}_{\text{Node2Vec}} \in R^{10743\times16}$, resulting in $\mathbf{X}_{\text{final}} \in R^{10743\times67}$.

2.3.4 Label Configuration

Binary labels were assigned based on known cancer driver gene catalogs:

- **Positive class (drivers)**: 796 genes from NCG v6.0, COSMIC CGC, and DigSEE
- **Negative class (non-drivers)**: 2,187 genes not annotated as drivers in NCG, COSMIC, OMIM, or KEGG cancer pathways
- **Unknown**: Remaining genes not in either set (excluded from training)

2.3.5 Cross-Validation Splits

The dataset was divided for ten repeats of five-fold cross-validation:

- Driver genes were randomly shuffled and divided into 5 equal chunks
- Non-driver genes were randomly shuffled and divided into 5 equal chunks
- For each fold, one chunk of drivers and one chunk of non-drivers formed the test set

- The remaining genes formed the training set
- This process was repeated 10 times with different random shuffles
- Final shape: $(10, 5, 2, 10743)$ where the third dimension represents [`train_mask`, `test_mask`]

3 Methods

3.1 Methods used for analysis

The TGCN architecture comprises two Transformer layers with weight sharing, residual enhanced activation and multi-head attention, followed by a Chebyshev GCN classifier [7, 8, 9]. The model was implemented in PyTorch and PyTorch Geometric, trained for 550 epochs per fold using Adam optimization with edge dropout ($p = 0.5$) and weight decay (5×10^{-4}).

PPI network analysis included degree, betweenness, closeness, PageRank, clustering coefficient and core number calculations using NetworkX. Degree distribution was fitted using the `powerlaw` package. Degree-preserving random networks were generated using edge swaps. Leiden community detection was performed on the largest connected component using igraph and leidenalg. Pathway enrichment was conducted using g:Profiler with custom background (1,933 scored genes) and FDR correction [14]. Jaccard pathway cross-talk was calculated for Reactome pathways with FDR < 0.05.

3.2 Statistical analysis

Model performance was evaluated using ROC-AUC and AUPR. Network properties were compared using Mann–Whitney U tests with Benjamini–Hochberg FDR correction. Spearman correlations assessed relationships between TGCN scores and network centrality. Fisher's exact test evaluated driver enrichment in Leiden modules. All analyses were performed in Python 3.13 with scikit-learn, scipy and statsmodels.

4 Results

4.1 Model performance and gene prioritization

The TGCN model achieved a mean ROC-AUC of 0.905 ± 0.015 and mean AUPR of 0.887 ± 0.018 across 25 cross-validation folds (Table 1).

Genes ranked with valid TGCN scores were ranked in descending order. The top 20 ranked genes included well-known cancer drivers such as TP53, CTNNB1, PIK3CA, KRAS, PTEN and EGFR (Figure 1), indicating that the model successfully recovers established cancer genes. Out-of-fold predictions were obtained for 1,933 of 10,743 genes, corresponding to the labelled training set.

4.2 PPI network topology of prioritized genes

The PPI network contained 10,743 nodes and 103,087 edges with 107 connected components. The top 100 TGCN-ranked genes exhibited significantly higher degree (median 55.5 vs 10.0, FDR $= 2.6 \times 10^{-30}$), betweenness (median 0.00118 vs 0.000036, FDR

Table 1: Model performance across cross-validation folds.

Metric	Mean	Standard Deviation
ROC-AUC	0.905	0.015
AUPR	0.887	0.018

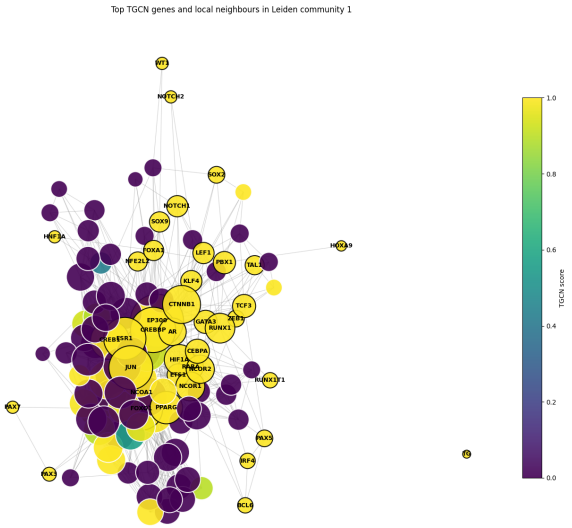


Figure 1: Top prioritized genes identified by TGCN framework

$= 7.5 \times 10^{-39}$) and closeness (median 0.285 vs 0.231, FDR $= 1.6 \times 10^{-42}$) compared to other scored genes (Table 2). TGCN scores were strongly correlated with closeness ($\rho = 0.67$, $p < 10^{-250}$) and betweenness ($\rho = 0.55$, $p < 10^{-150}$), indicating that prioritized genes occupy central network positions.

Table 2: Network properties of top 100 TGCN genes versus other scored genes.

Property	Top 100 Median	Other Median	FDR
Degree	55.5	10.0	2.6×10^{-30}
Betweenness	0.00118	0.000036	7.5×10^{-39}
Closeness	0.285	0.231	1.6×10^{-42}
PageRank	0.000267	0.000074	2.5×10^{-36}
Core number	21.0	8.0	1.2×10^{-20}
Clustering	0.165	0.374	9.7×10^{-13}

4.3 Network Correlation

Spearman rank correlation was calculated between TGCN score and network property, the result is in the table below. We can interpret that the genes that TGCN predict have greater network centrality.

Table 3: Spearman correlations between TGCN scores and network properties.

Network property	Spearman (ρ)	Interpretation
Degree	0.480	Moderate positive association
Betweenness	0.553	Moderate positive association
Closeness	0.671	Strong positive association
Eigenvector	0.403	Moderate positive association
PageRank	0.457	Moderate positive association
Clustering	-0.211	Weak negative association
Core number	0.423	Moderate positive association

4.4 Heavy-tailed degree distribution

The degree distribution exhibited a heavy-tailed structure with maximum degree 366 and median degree 9 (Figure 2). Power-law fitting yielded $\alpha = 2.42$ for $k \geq 25$, but the power-law model was significantly less likely than a lognormal model (likelihood ratio $= -89.1$, $p = 7.0 \times 10^{-18}$). This indicates that the network is heavy-tailed but not strictly scale-free, consistent with biological networks that contain hubs but deviate from pure power-law behaviour.

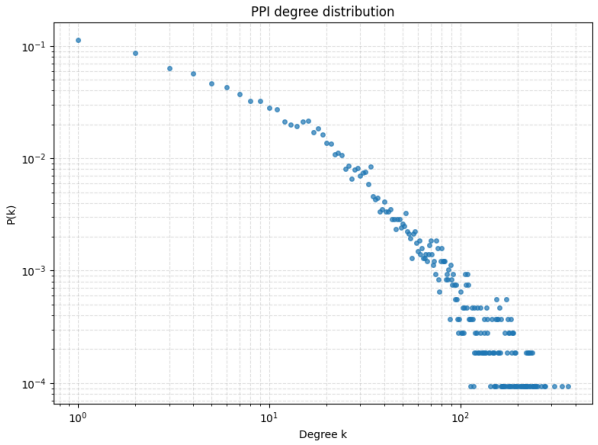


Figure 2: PPI degree distribution.

4.5 Degree-preserving random networks

Compared with 20 degree-preserving random networks, the observed network showed substantially higher average clustering (0.422 vs 0.018) and strong positive degree assortativity (0.449 vs -0.019) (Table 4). This demonstrates that the network contains biologically meaningful modules and hub–hub connectivity beyond what is expected from degree distribution alone.

4.6 Leiden community detection

Leiden algorithm identified 33 communities in the largest connected component (10,493 nodes, 102,934 edges). Four communities were significantly enriched for known cancer drivers (FDR < 0.05), with odds ratios between 1.69 and 2.14 (Table 5). Community

Table 4: Comparison of observed network with degree-preserving random networks.

Property	Observed	Random Mean ± SD
Average clustering	0.422	0.018 ± 0.001
Degree assortativity	0.449	-0.019 ± 0.002

10 showed the highest driver fraction (13.8%) and odds ratio (2.14), indicating strong functional coherence. The top 100 TGCN genes were concentrated in communities 0, 1, 3 and 10, which together contained 71 of the top 100 genes.

Table 5: Driver-enriched Leiden communities.

Community	Size	Drivers	Driver Fraction	Odds Ratio	FDR
1	1,274	154	12.1%	1.96	6.6×10^{-10}
3	1,039	122	11.7%	1.84	3.8×10^{-7}
0	1,357	147	10.8%	1.69	1.4×10^{-6}
10	421	58	13.8%	2.14	1.0×10^{-5}

4.7 Network perturbation

Removal of the top 100 TGCN genes increased the number of connected components from 12 to 43 and reduced the largest component size from 10,379 to 10,337 (Table 6). Random removal of 100 genes resulted in 12.3 ± 4.3 components on average. This indicates that TGCN-prioritized genes are more central in the network than average genes, although degree-matched controls are required to determine whether this effect is due to TGCN prioritization or simply network centrality.

Table 6: Network perturbation by gene removal.

Removal Type	Components	Largest Component Size
Top 100 TGCN	43	10,337
Random (mean ± SD)	12.3 ± 4.3	$10{,}379 \pm 7$

4.8 Pathway enrichment

Reactome enrichment of the top 100 TGCN genes revealed 368 significant pathways (FDR < 0.05). The most significant pathways included Developmental Biology ($p = 1.0 \times 10^{-22}$), Diseases of signal transduction by growth factor receptors ($p = 1.1 \times 10^{-18}$), Generic Transcription Pathway ($p = 1.8 \times 10^{-16}$), Signal Transduction ($p = 2.1 \times 10^{-15}$) and PI3K/AKT Signaling in Cancer ($p = 1.2 \times 10^{-9}$) (Figure 3). The top 100 candidate genes (label = 0) were enriched for DNA Replication ($p = 3.6 \times 10^{-4}$), G1/S Transition ($p = 7.6 \times 10^{-3}$) and Cell Cycle ($p = 2.7 \times 10^{-2}$), indicating involvement in fundamental cellular processes.

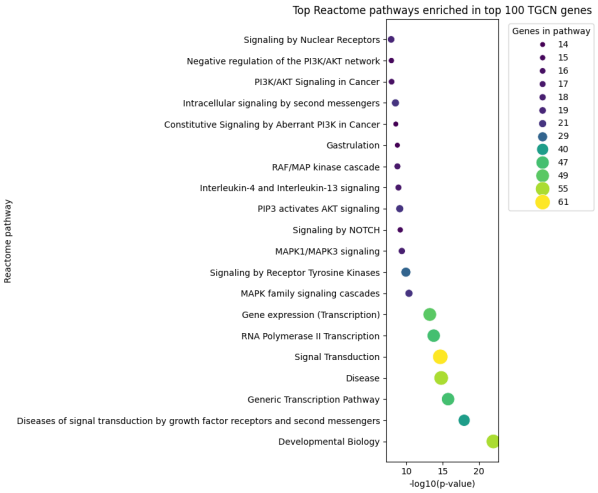


Figure 3: Top Reactome pathways enriched in top 100 TGCN genes. Dot size represents number of genes in pathway, colour represents significance.

4.9 Jaccard pathway cross-talk

Among 368 significant Reactome pathways, 5,079 pathway pairs showed Jaccard similarity ≥ 0.30, indicating substantial gene overlap. Many highly similar pairs corresponded to hierarchical or mutant-specific annotations, such as "Signaling by NOTCH1 in Cancer" and "Constitutive Signaling by NOTCH1 PEST Domain Mutants in Cancer" ($J = 1.0$, 6 shared genes), or "Generic Transcription Pathway" and "RNA Polymerase II Transcription" ($J = 1.0$, 47 shared genes). This indicates that TGCN-prioritized genes are concentrated in a small number of core biological processes, and that many Reactome pathways represent different annotations of the same underlying gene sets.

5 Discussion and Conclusion

This study demonstrates that a multi-view TGCN model can prioritize biologically meaningful cancer driver genes. The top-ranked genes occupy central positions in the PPI network, are concentrated in driver-enriched network modules, and are enriched for cancer-relevant pathways including growth-factor signalling, transcriptional regulation and PI3K–AKT signalling.

Our results support previous findings that cancer driver genes tend to be network hubs and participate in core cellular processes [11, 12]. The strong enrichment for developmental biology and signal transduction pathways corroborates the established role of these processes in cancer progression [1, 3]. The candidate gene enrichment for DNA replication and cell-cycle pathways suggests that the model identifies genes involved in fundamental cellular processes even when they are not included in reference driver sets.

We propose the following model: TGCN prioritizes genes that (i) occupy central network positions, (ii) participate in coherent functional modules, and (iii) are involved in core oncogenic signalling pathways. This model is consistent with the network-based view of cancer, which posits that driver genes perturb cellular networks rather than acting in isolation [13, 10].

Taken together, the results consistently show that TGCN-prioritized genes are more central, more modular and more pathway-enriched than expected by chance, indicating

that the model captures biologically meaningful patterns. Future studies should validate these predictions experimentally, extend the analysis to tissue-specific networks, and incorporate additional data types such as expression, phosphorylation and structural information.

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